

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**



Lecture No.- 03

By- Rahul Sir

Topics to be Covered

Topic



- 1:- Multiplexer ✓
- 2:- Types of Multiplexer ✓
- 3:- 2x1 mux, 4x1 mux , 8x1 mux, 16x1 mux ✓
- 4:- 8x1 mux by using 2x1 mux ✓
- 5:- 8x1 mux by using 4x1 mux – special case
(Enabling System)
- 6:- 16x1 mux using 4x1 demux
- 7:- 16x1 mux using 2x1 demux
- 8:- Demultiplexer ✓



Applications of Full Subtractor

The following are some important applications of full subtractor –

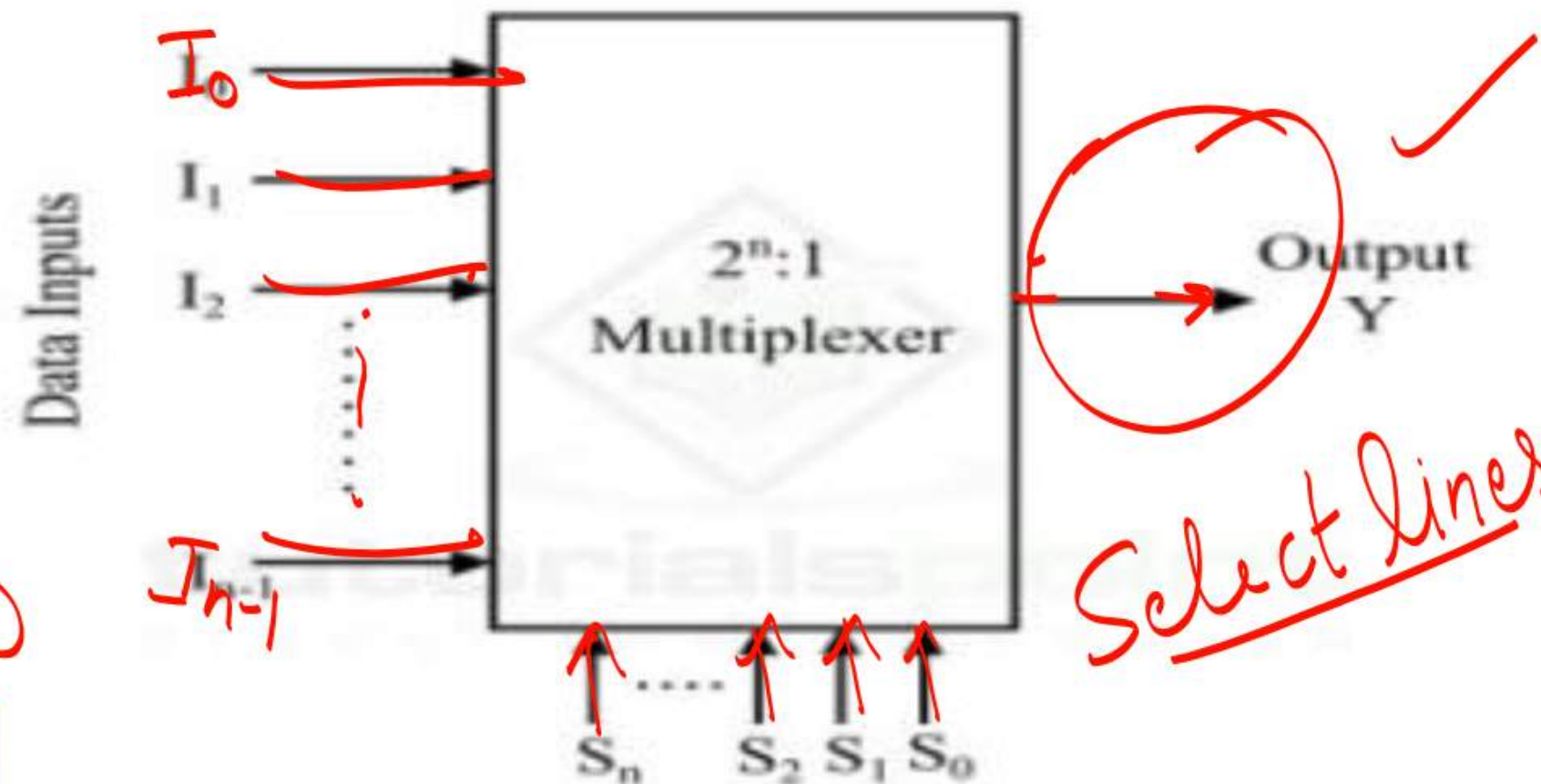
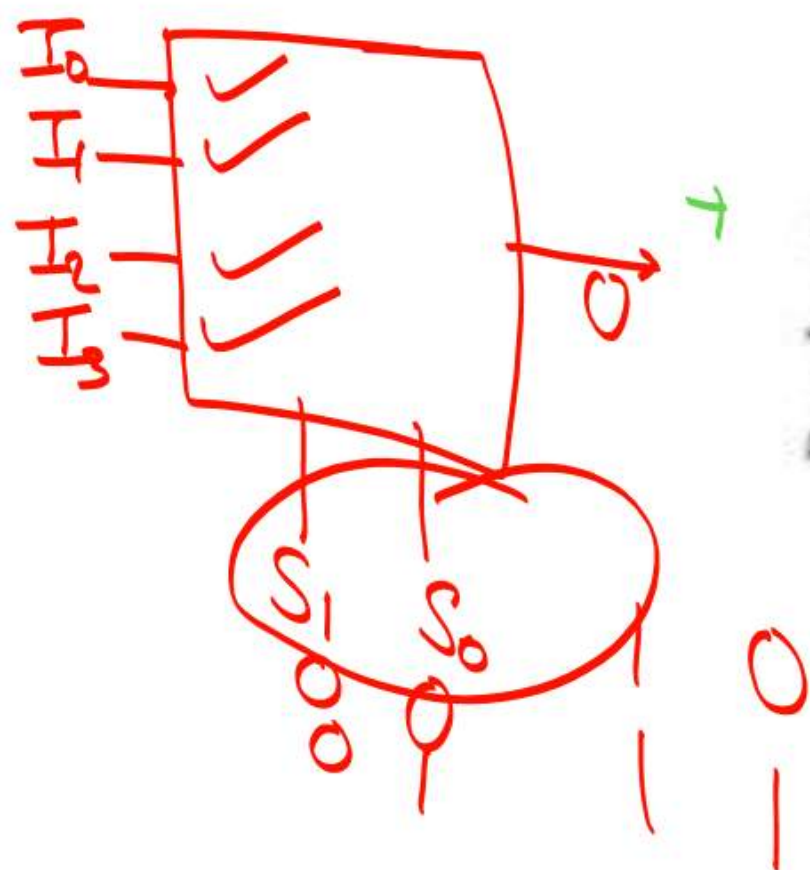
- Full subtractors are used in ALU (Arithmetic Logic Unit) in computers CPUs.
- Full subtractors are extensively used to perform arithmetical operations like subtraction in electronic calculators and many other digital devices.
- Full subtractors are used in different microcontrollers for arithmetic subtraction.
- They are used in timers and program counters (PC).
- Full subtractors are also used in processors to compute addresses, tables, etc.
- Full subtractors are also used in DSP (Digital Signal Processing) and networking based systems.



What is a Multiplexer?

As already mentioned, a **multiplexer**, also referred to as **MUX**, is a combination logic circuit that is designed to accept multiple input signals and transfer only one of them through the output line. In simple words, a multiplexer is a digital logic device that selects one-out-of-N ($N = 2^n$) input data sources and transmits the selected data to a single output line.

The multiplexer is also called **data selector** as it selects one from several. The block diagram of a typical $2^n:1$ multiplexer is shown in Figure 1.



Input Combination = 2^n
No of Select line = n

Select lines



In the case of multiplexer, the selection of desired data input to flow through the output line is controlled with the help of **SELECT lines**. In the block diagram of mux in Figure 1, $I_0, I_1, \dots, I_{n-1}$, i.e., (2^n) are the input lines, and "**n**" be the select lines. These select lines will determine which input is to be routed to the output.

Hence, the multiplexer works as a multi-position switch whose operation is controlled by digital signals. These digital control signals are applied to the select lines to determine which data input will be switched to the output line.



Function of Multiplexer

Multiplexer is a digital logic device which is used to perform **multiplexing** of data. Where, multiplexing simply means sharing of data. Technically, when a particular data is selected from multiple input data sources and transmitted the selected data to a single output channel, it is called **multiplexing**.

There are two types of multiplexing namely, frequency multiplexing and time multiplexing.

When multiple devices are connected to a single transmission line in a system. At any point of time, only one device is using the line to transmit data, then this is called time multiplexing. On the other hand, when multiple devices share a common line to transmit data but at different frequencies, it is called frequency multiplexing.



1:- 2x1 Multiplexer ✓

2:- 4x1 Multiplexer ✓

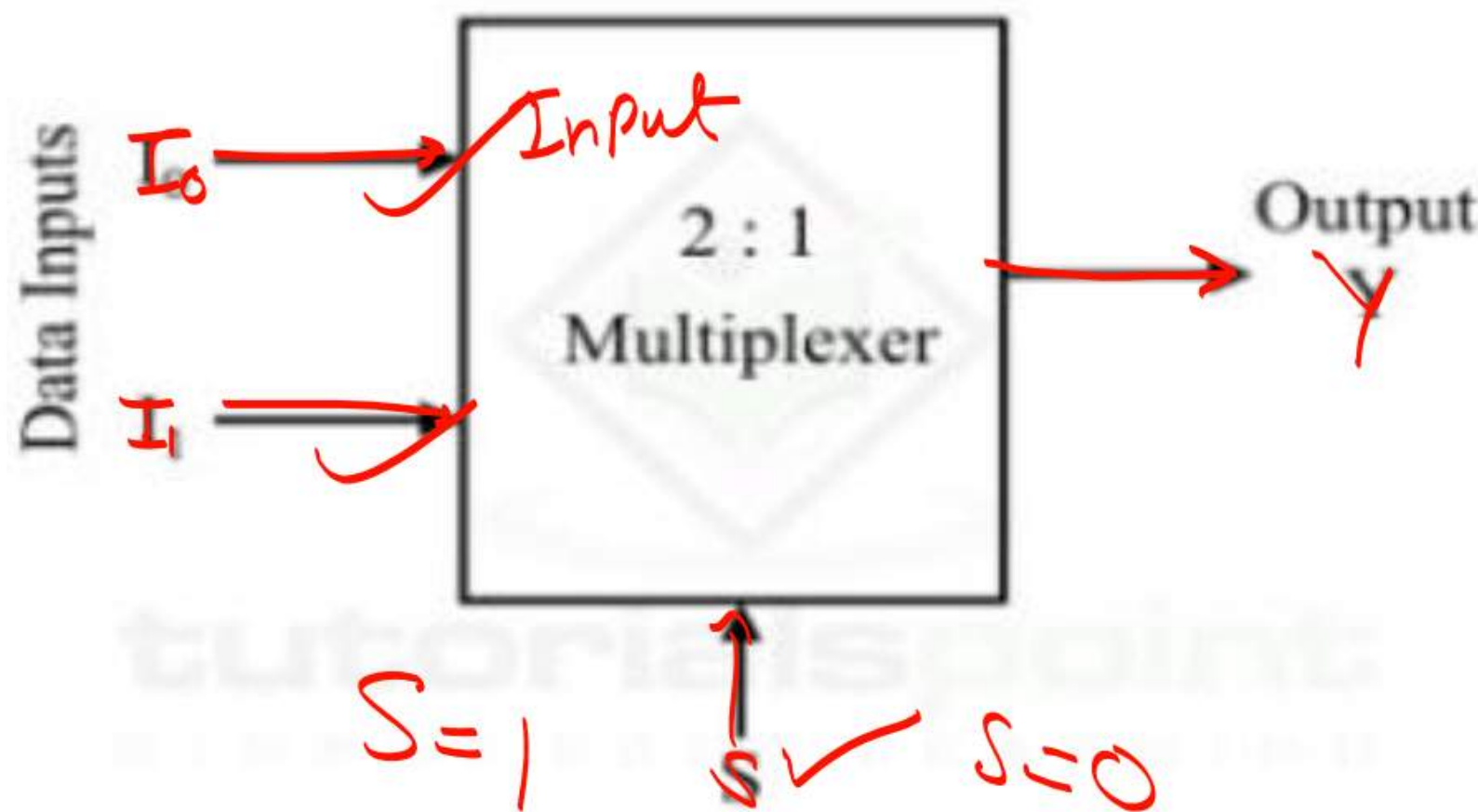
3:- 8x1 Multiplexer ✓

4:- 16x1 Multiplexer ✓



2x1 Multiplexer

The block diagram of a 2x1 multiplexer is shown in Figure 2. The 2x1 multiplexer is basic two input multiplexer which has two data input lines designated as I_0 and I_1 , one data select line denoted by S and one output line denoted by Y . The 2x1 mux is used to connect two 1-bit data sources to a common designation.



Output depend
① Select line
② Input (Selected)



Topic : 2x1 Multiplexer



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In the 2×1 multiplexer, the logic level of the digital signal applied to the select line S determines which data input will pass through the output line. The operation of the 2×1 multiplexer can be understood from the following truth table.

Select Line (S)	Output (Y)
0 → S'	I ₀
1 → S	I ₁

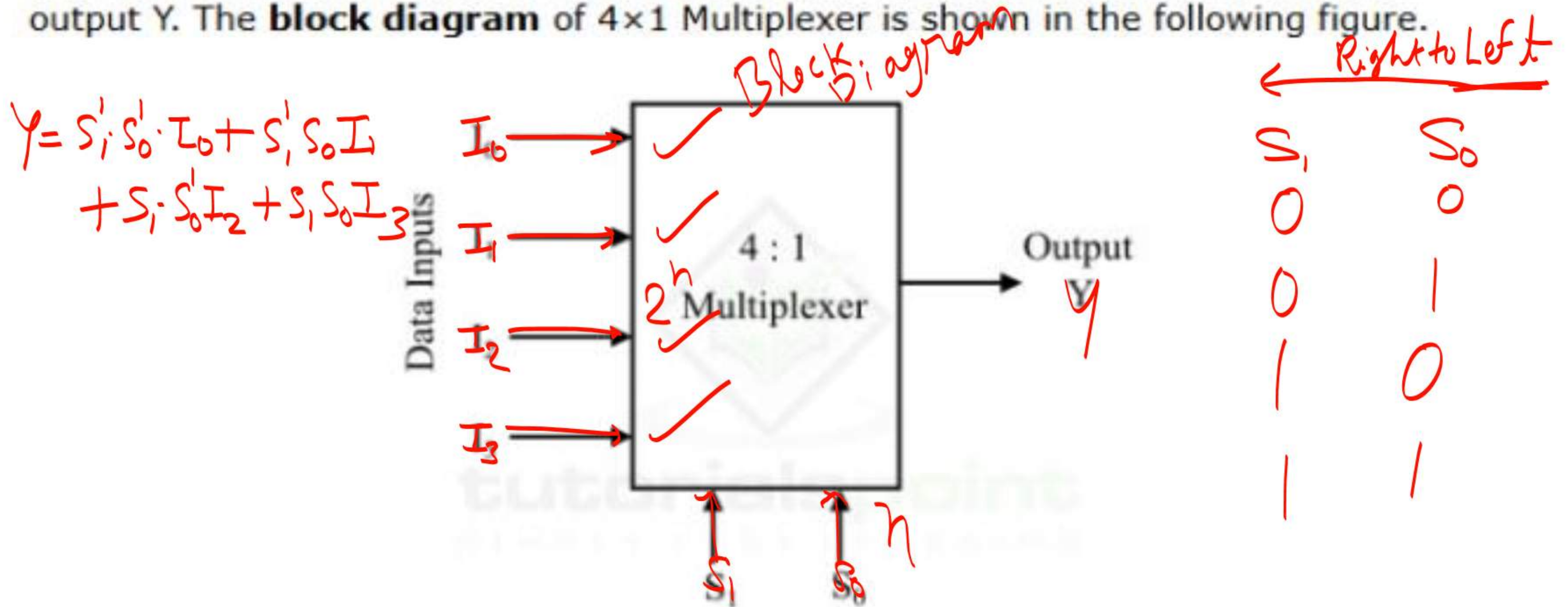
$$Y = S' \cdot I_0 + S \cdot I_1$$

SOP $\boxed{Y = S' \cdot I_0 + S \cdot I_1}$



4x1 Multiplexer

4x1 Multiplexer has four data inputs I_3, I_2, I_1 & I_0 , two selection lines s_1 & s_0 and one output Y . The **block diagram** of 4x1 Multiplexer is shown in the following figure.





Topic : 4x1 Multiplexer



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One of these 4 inputs will be connected to the output based on the combination of inputs present at these two selection lines. **Truth table** of 4×1 Multiplexer is shown below.

Selection Lines		Output
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

From Truth table, we can directly write the **Boolean function** for output, Y as

$$Y = S_1'S_0'I_0 + S_1'S_0I_1 + S_1S_0'I_2 + S_1S_0I_3$$



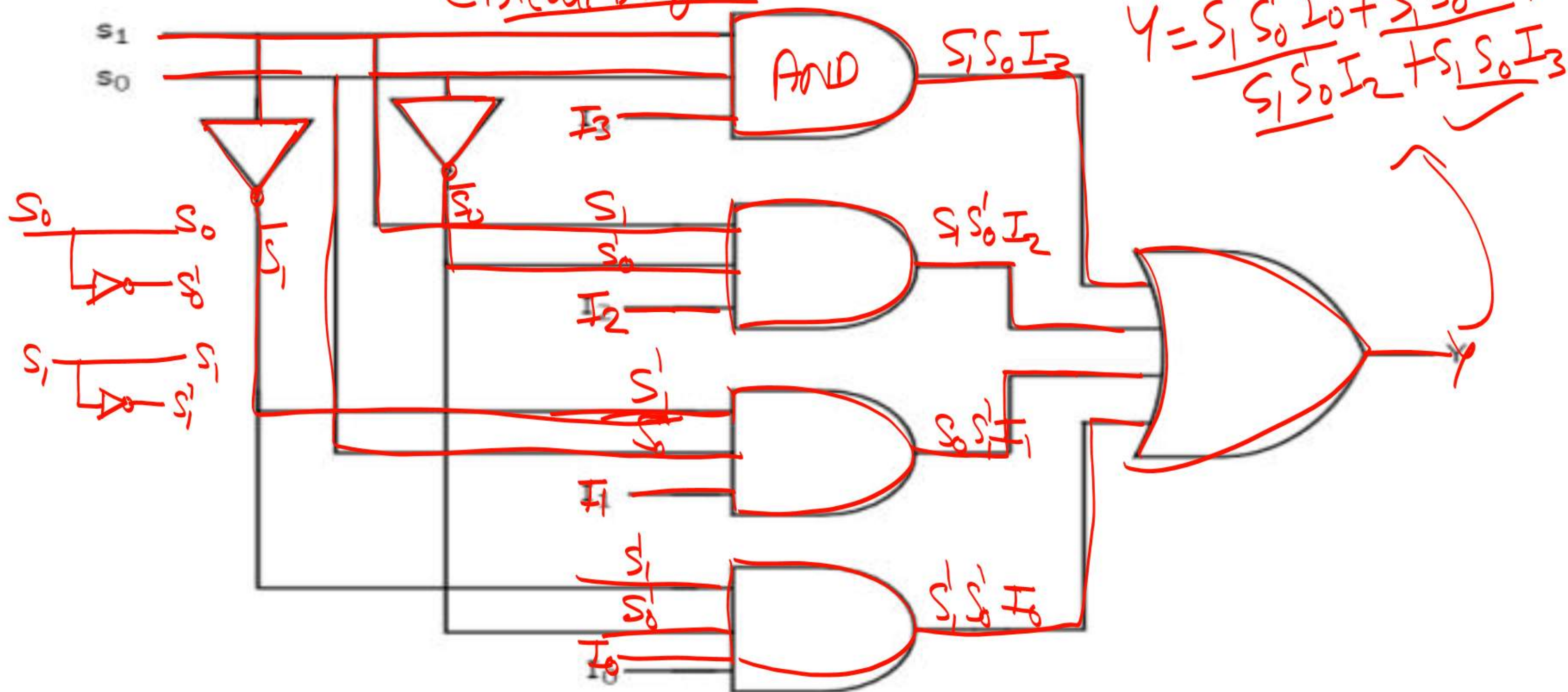
Topic : 4x1 Multiplexer



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We can implement this Boolean function using Inverters, AND gates & OR gate. The **circuit diagram** of 4x1 multiplexer is shown in the following figure.

Circuit Diagram





8x1 Multiplexer

In this section, let us implement 8x1 Multiplexer using 4x1 Multiplexers and 2x1 Multiplexer. We know that 4x1 Multiplexer has 4 data inputs, 2 selection lines and one output. Whereas, 8x1 Multiplexer has 8 data inputs, 3 selection lines and one output.

So, we require two **4x1 Multiplexers** in first stage in order to get the 8 data inputs. Since, each 4x1 Multiplexer produces one output, we require a **2x1 Multiplexer** in second stage by considering the outputs of first stage as inputs and to produce the final output.



Topic : 8x1 Multiplexer



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Let the 8x1 Multiplexer has eight data inputs I_7 to I_0 , three selection lines s_2 , s_1 & s_0 and one output Y . The **Truth table** of 8x1 Multiplexer is shown below.

Selection Inputs			Output
S_2	S_1	S_0	Y
0	0	0	I_0
1	0	1	I_1
2	1	0	I_2
3	1	1	I_3
4	0	0	I_4
5	0	1	I_5
6	1	0	I_6
7	1	1	I_7



Topic : 8x1 Multiplexer

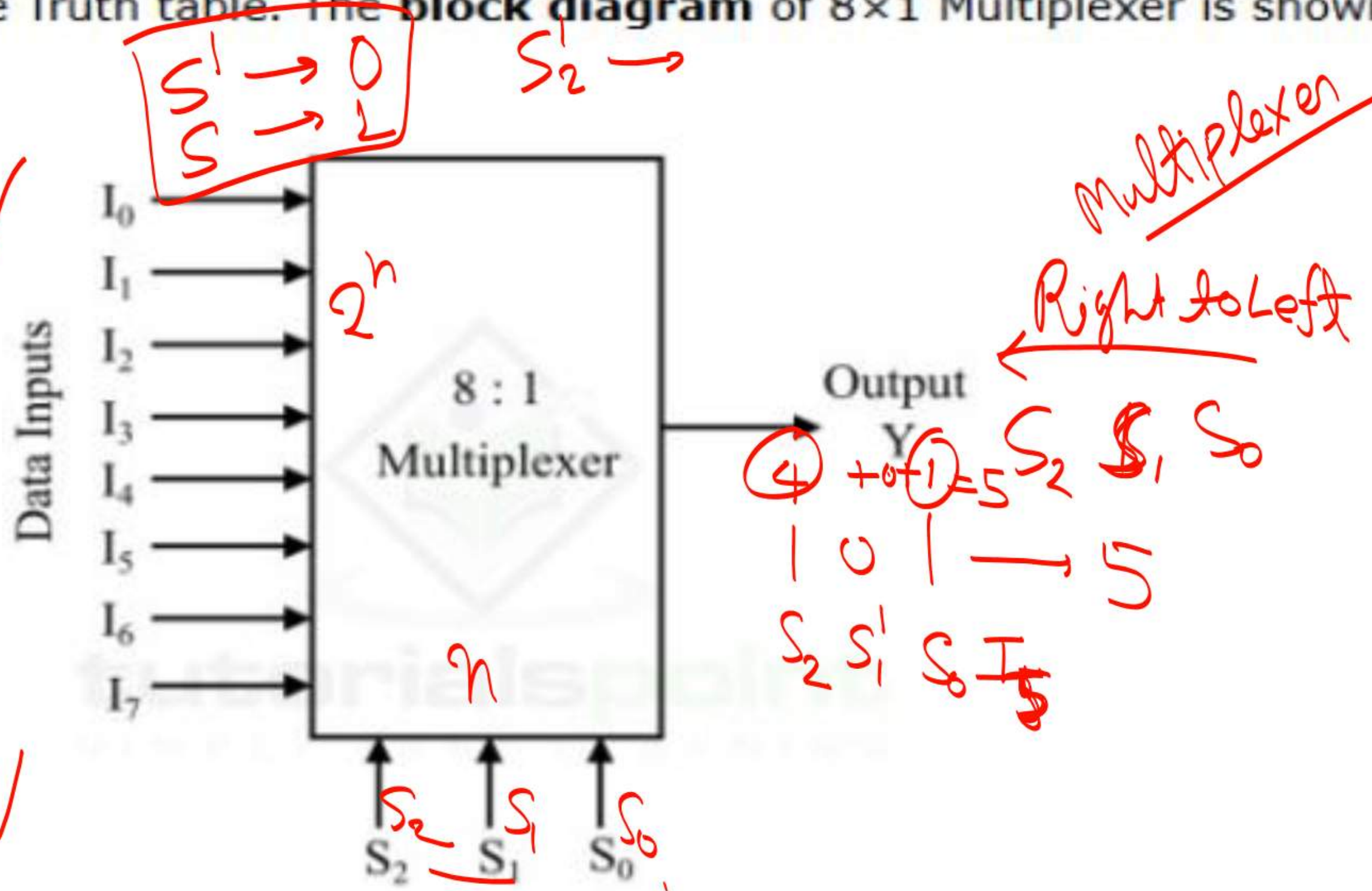


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We can implement 8x1 Multiplexer using lower order Multiplexers easily by considering the above Truth table. The **block diagram** of 8x1 Multiplexer is shown in the following figure.

Handwritten Boolean expression for the output Y of an 8x1 Multiplexer:

$$Y = S_2' S_1' S_0' I_0 + S_2' S_1' S_0 I_1 + S_2' S_1 S_0' I_2 + S_2' S_1 S_0 I_3 + S_2 S_1' S_0' I_4 + S_2 S_1' S_0 I_5 + S_2 S_1 S_0' I_6 + S_2 S_1 S_0 I_7$$

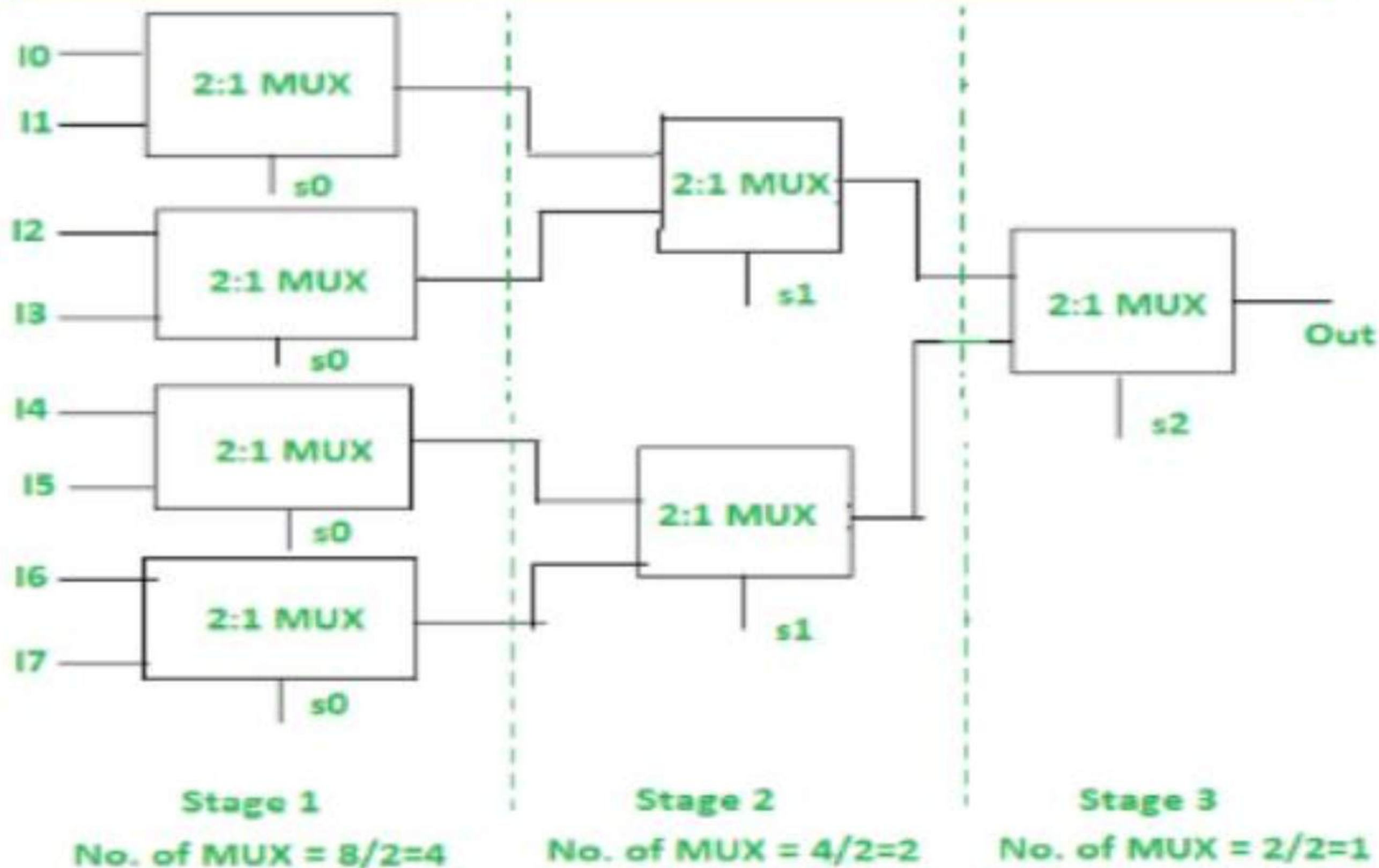




Topic : 8x1 Multiplexer by using 2 x1 mux



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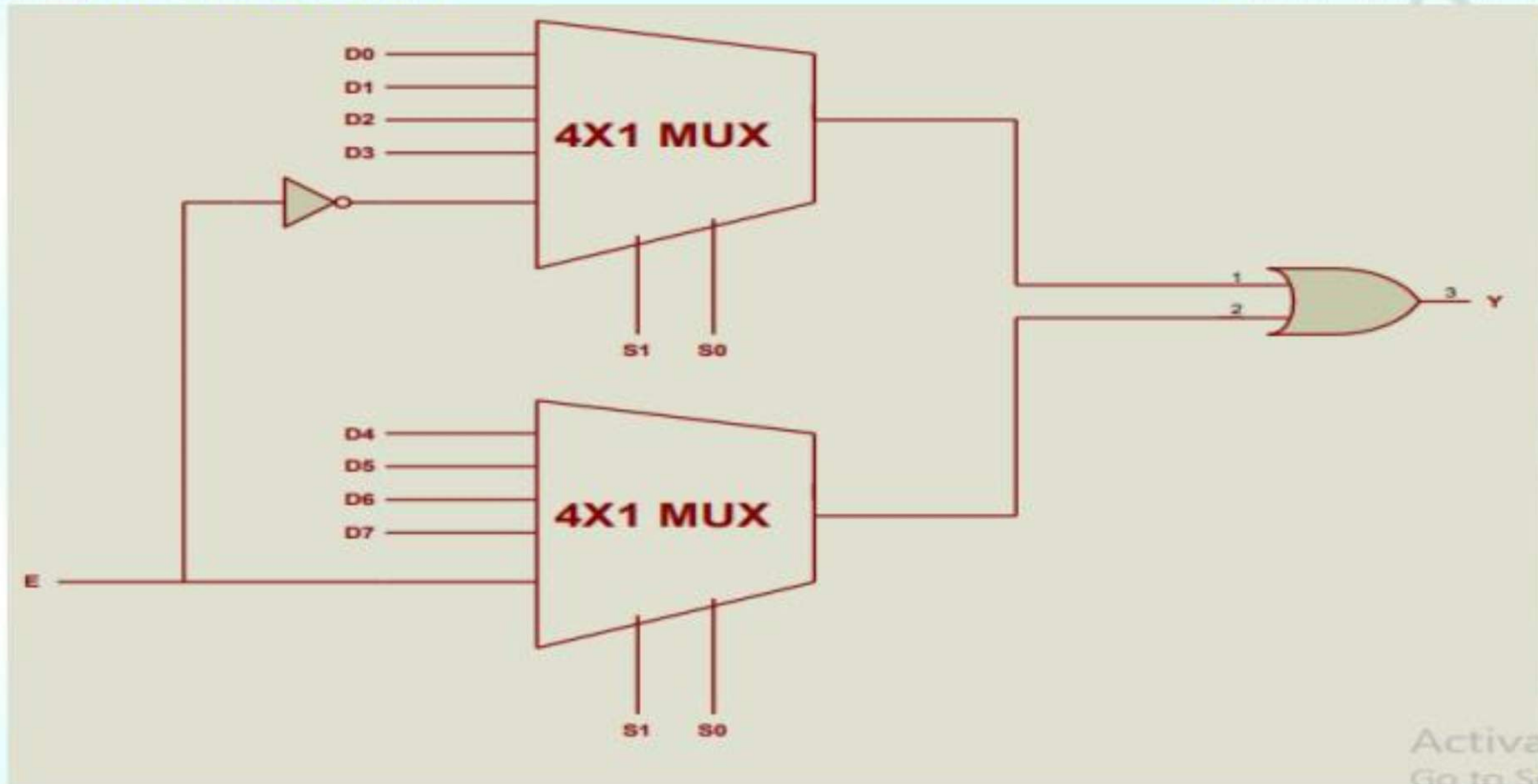




Topic : 8x1 Multiplexer by using 4x1 Mux- special case (Enable System)



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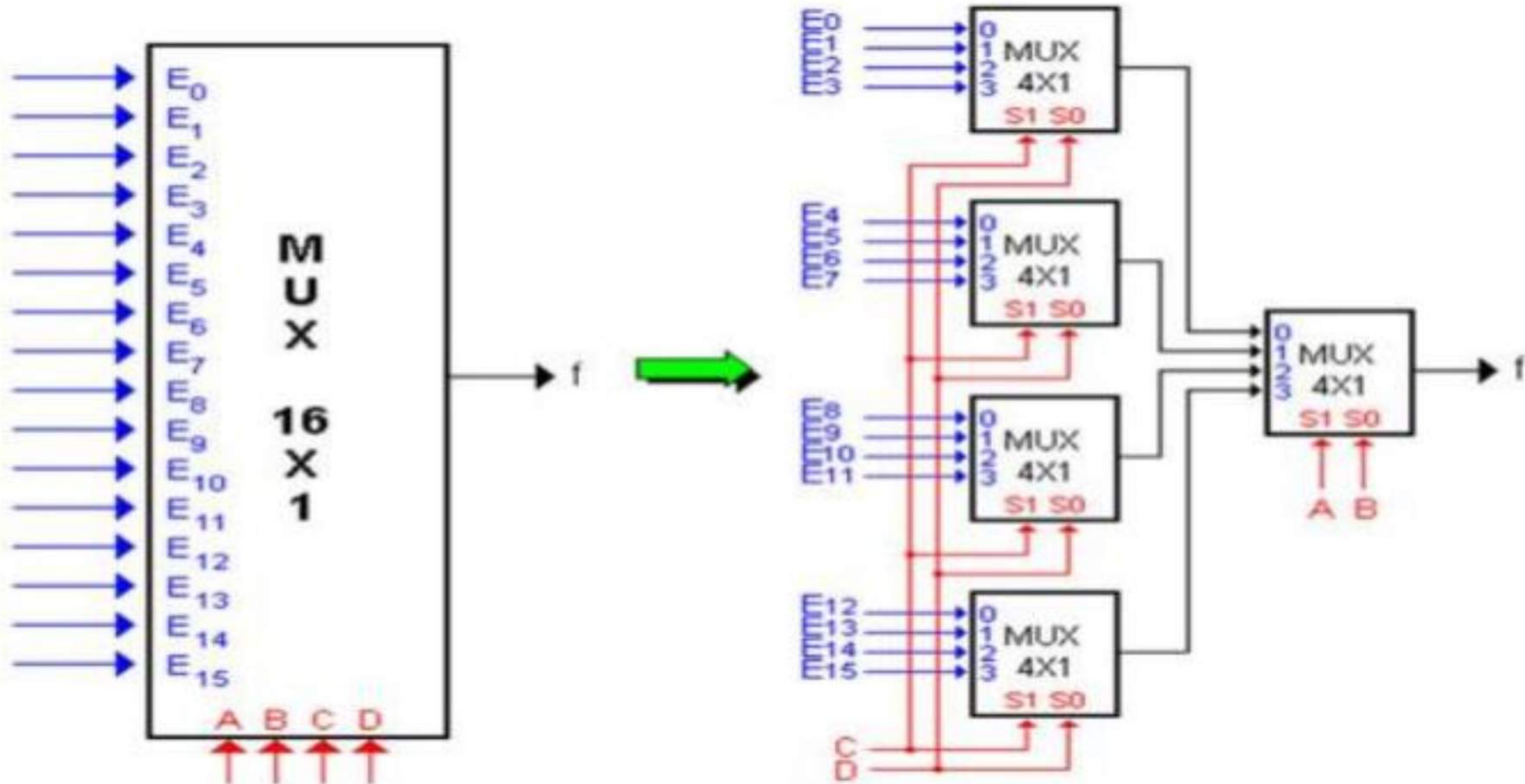




Topic : 16x1 Multiplexer by using 4x1 Mux



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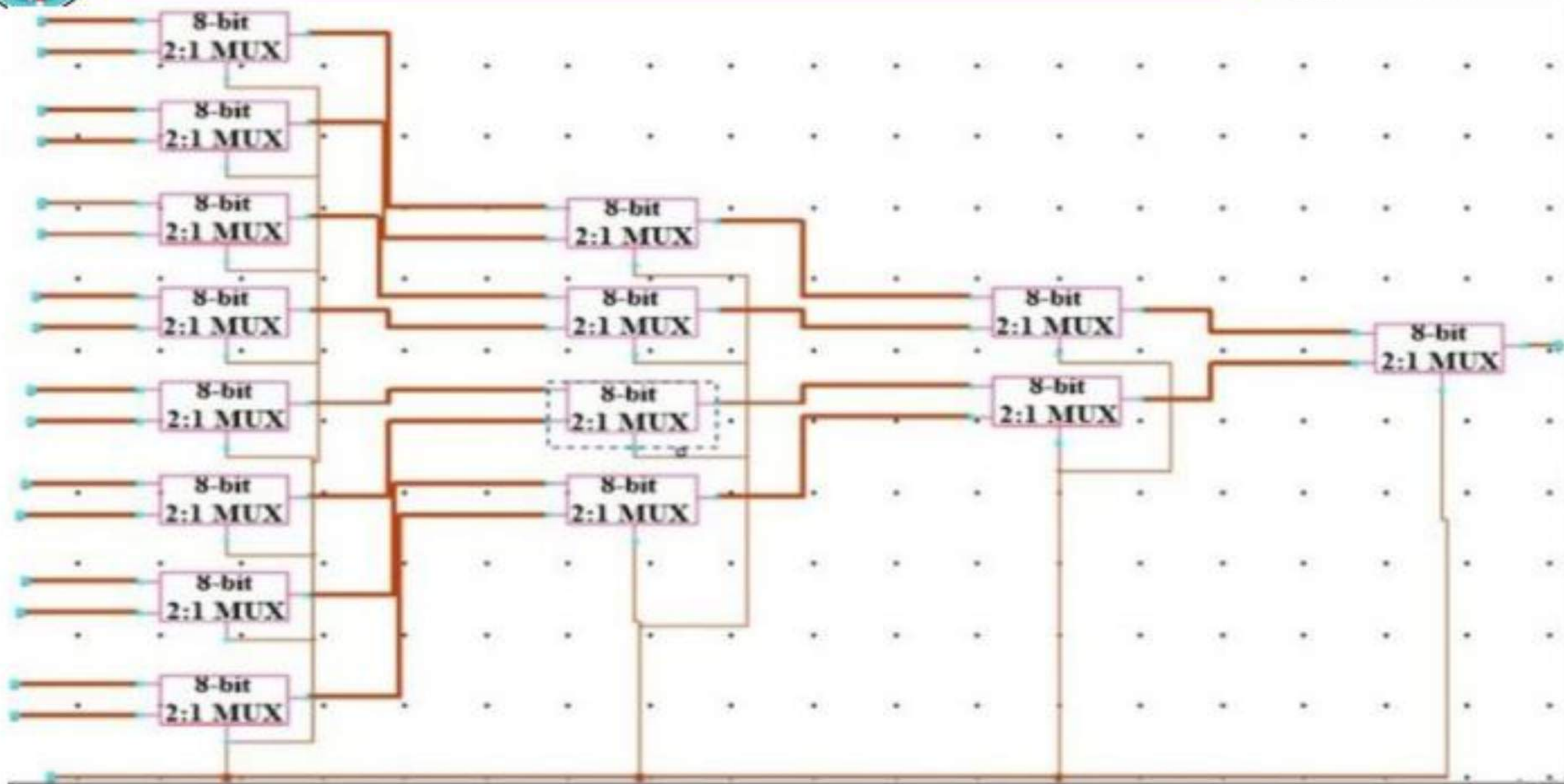




Topic : 16x1 Multiplexer by using 2x1 Mux



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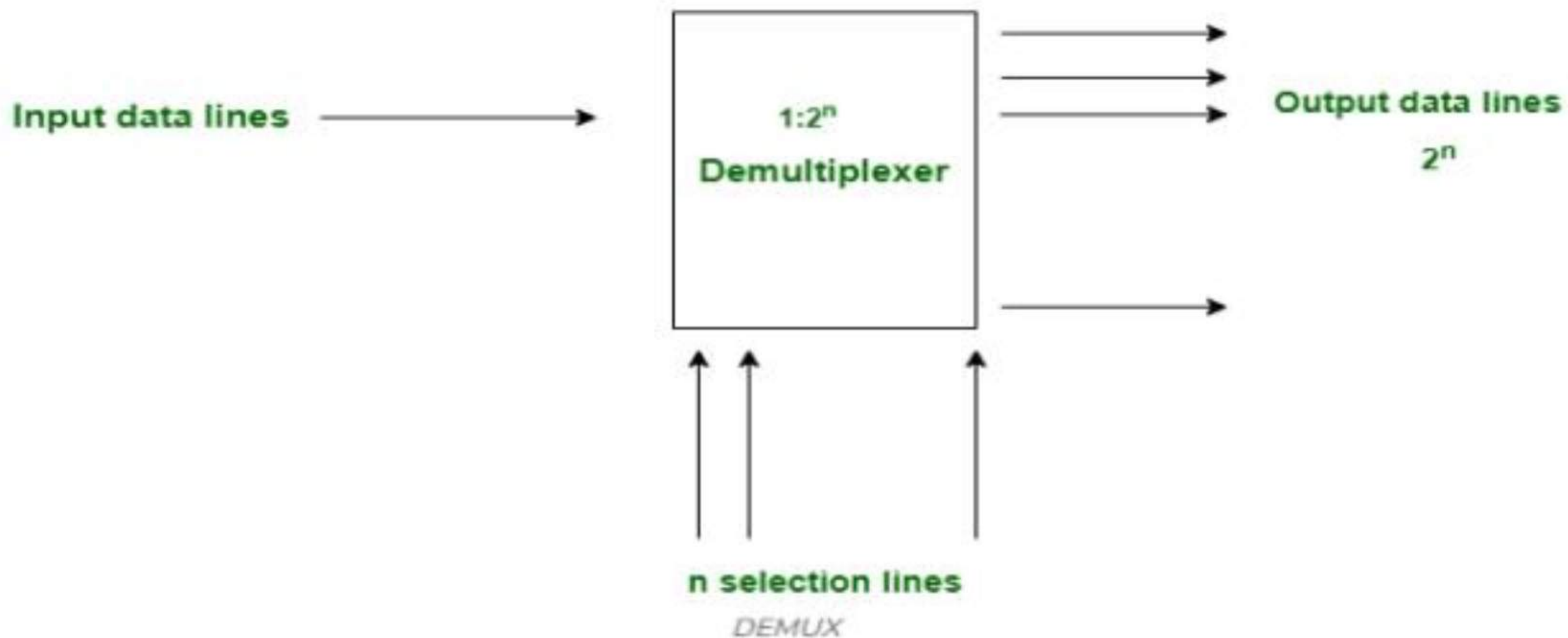
What is a DEMUX?

The DEMUX is a digital information processor. It takes input from one source and also converts the data to transmit towards various sources. The demultiplexer has one data input line. The demultiplexer has several control lines (also known as select lines). These lines determine to which output the input data should be sent. The number of control lines determines the number of output lines.



General Block Diagram Of A DEMUX

Here is the basic block diagram of a DEMUX as mentioned below.

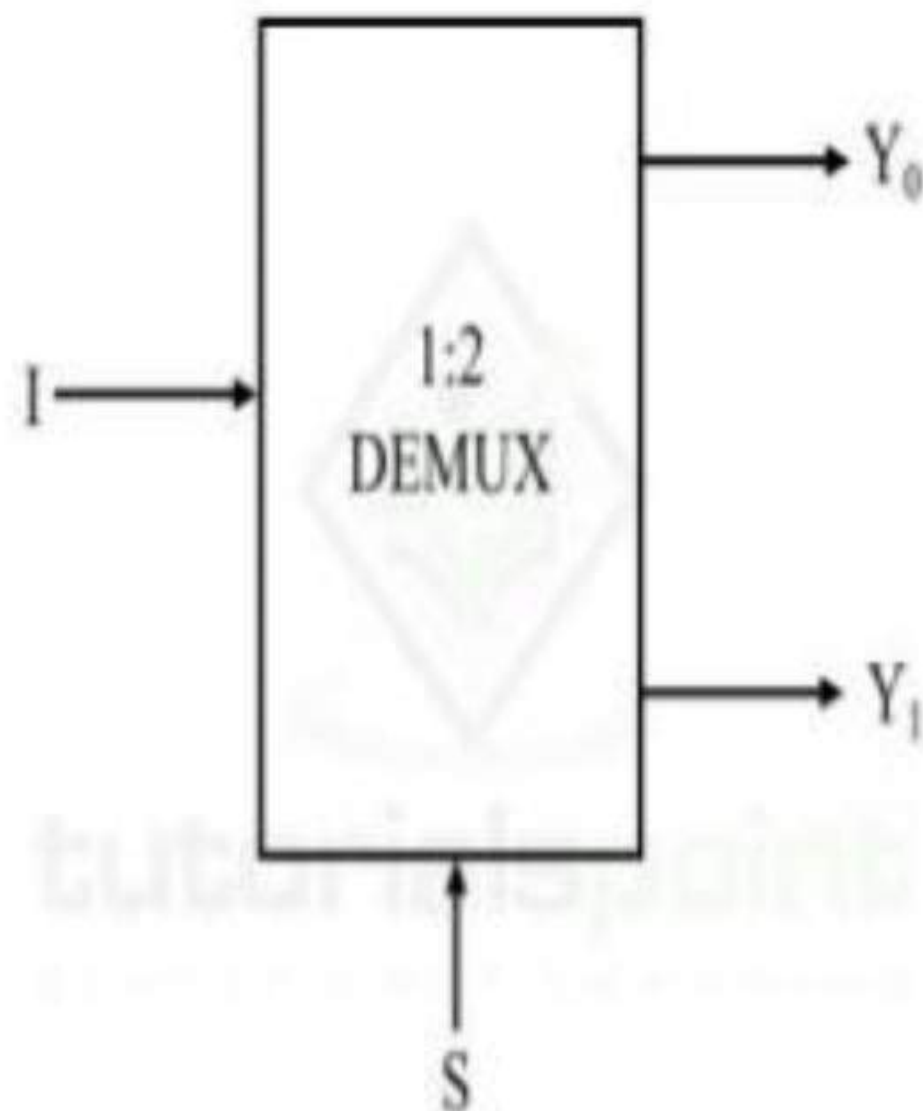




1x2 Demultiplexer

The operation of the 1x2 Demultiplexer can be analyzed with the help of its function table given below.

The functional block diagram of a 1x2 Demultiplexer is shown in



Select Line	Outputs	
S	Y_1	Y_0
0	0	I
1	I	0

From this function table of 1x2 Demultiplexer, we can directly derive the Boolean expression for each output as follow.

$$Y_0 = \bar{S} I$$

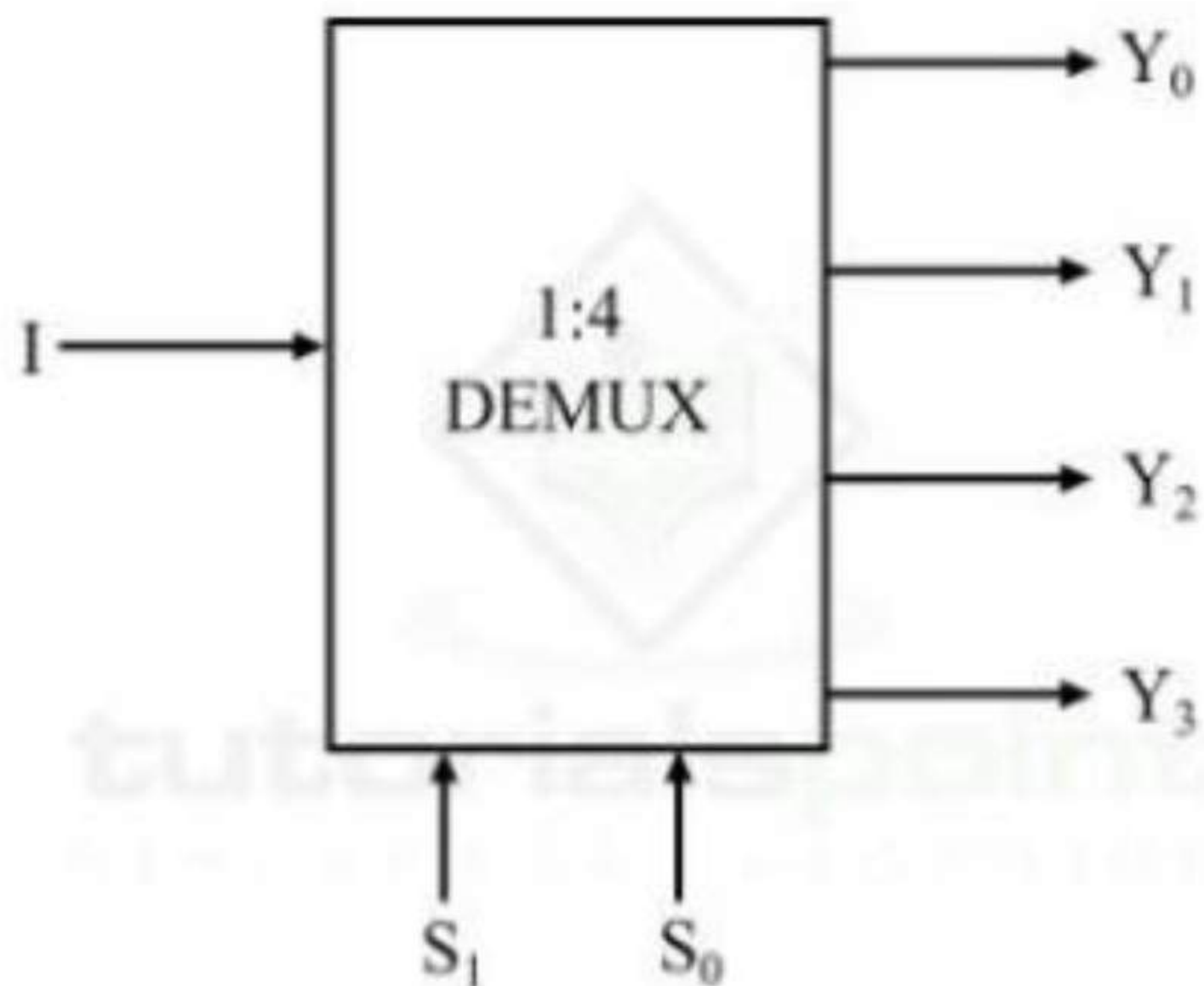
And,

$$Y_1 = S I$$



1x4 Demultiplexer

The functional block diagram of 1x4 Demultiplexer is shown in Figure-3.





Topic : 1x4 Dmux



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The operation of the 1×4 Demultiplexer can be understood with the help of its function table given below.

Select Line		Outputs			
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

From this truth table of 1×4 Demultiplexer, we can directly write the Boolean expression for each output as follow.

$$Y_0 = \bar{S}_1 \bar{S}_0 I$$

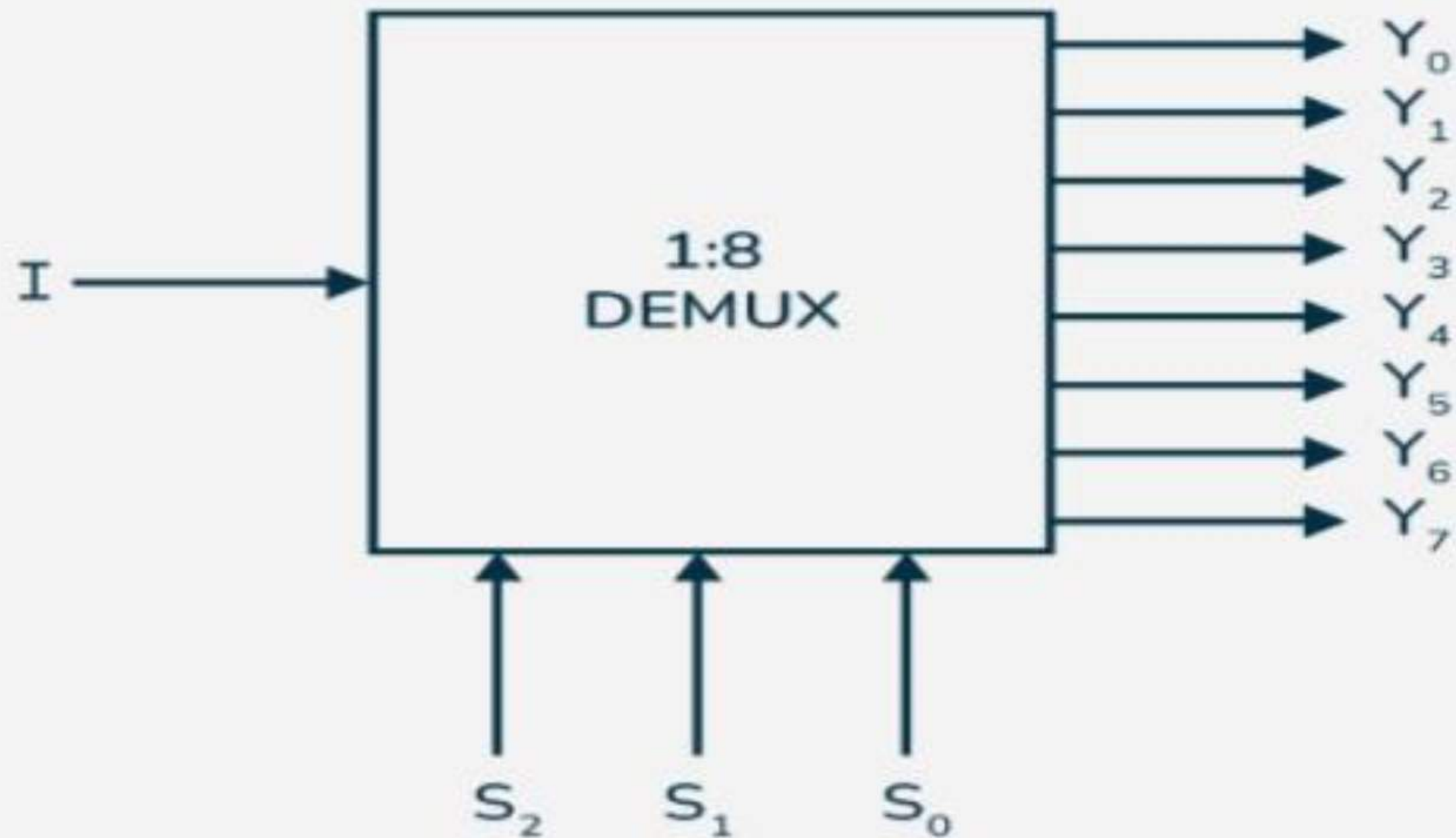
$$Y_1 = \bar{S}_1 S_0 I$$

$$Y_2 = S_1 \bar{S}_0 I$$

$$Y_3 = S_1 S_0 I$$



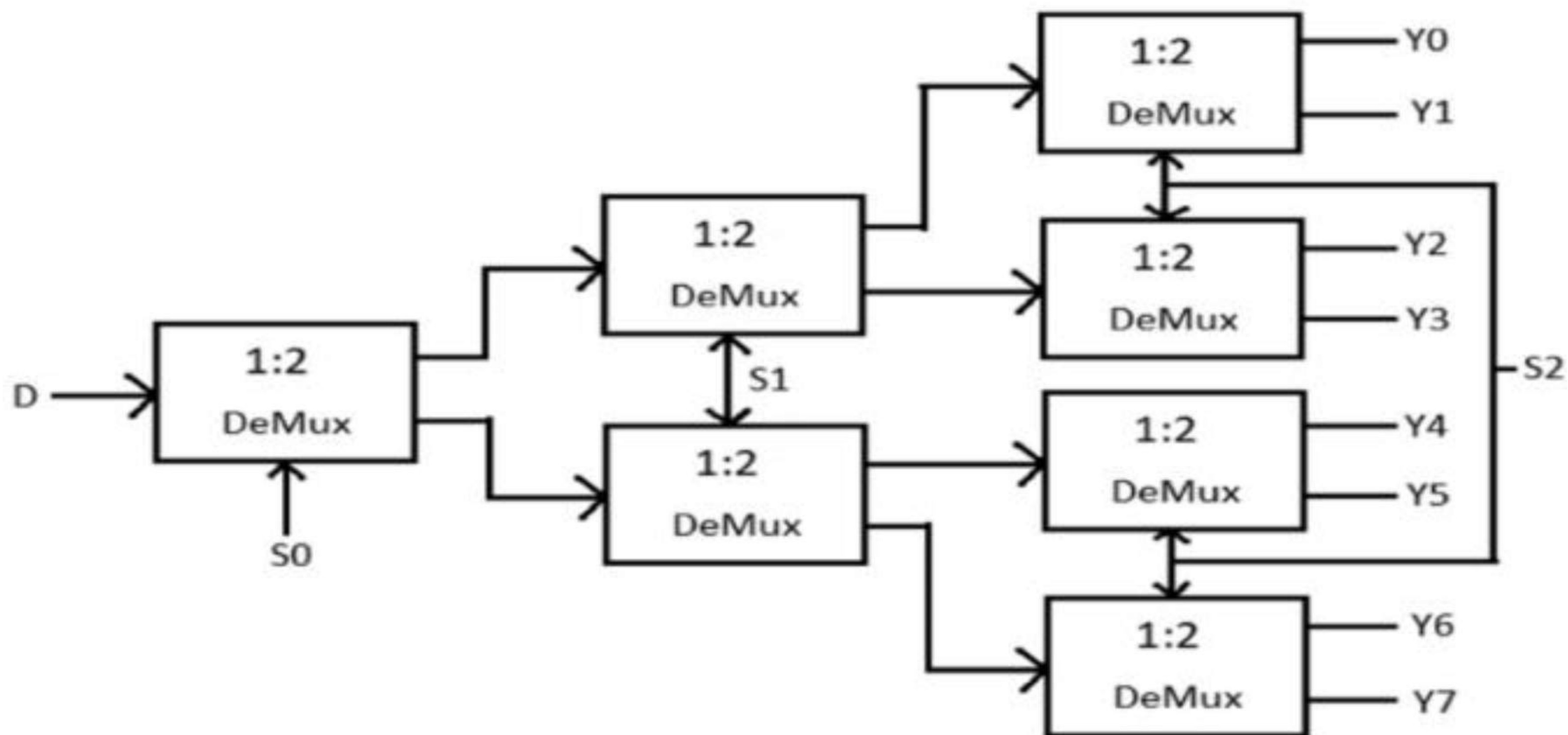
1:8 Demultiplexer

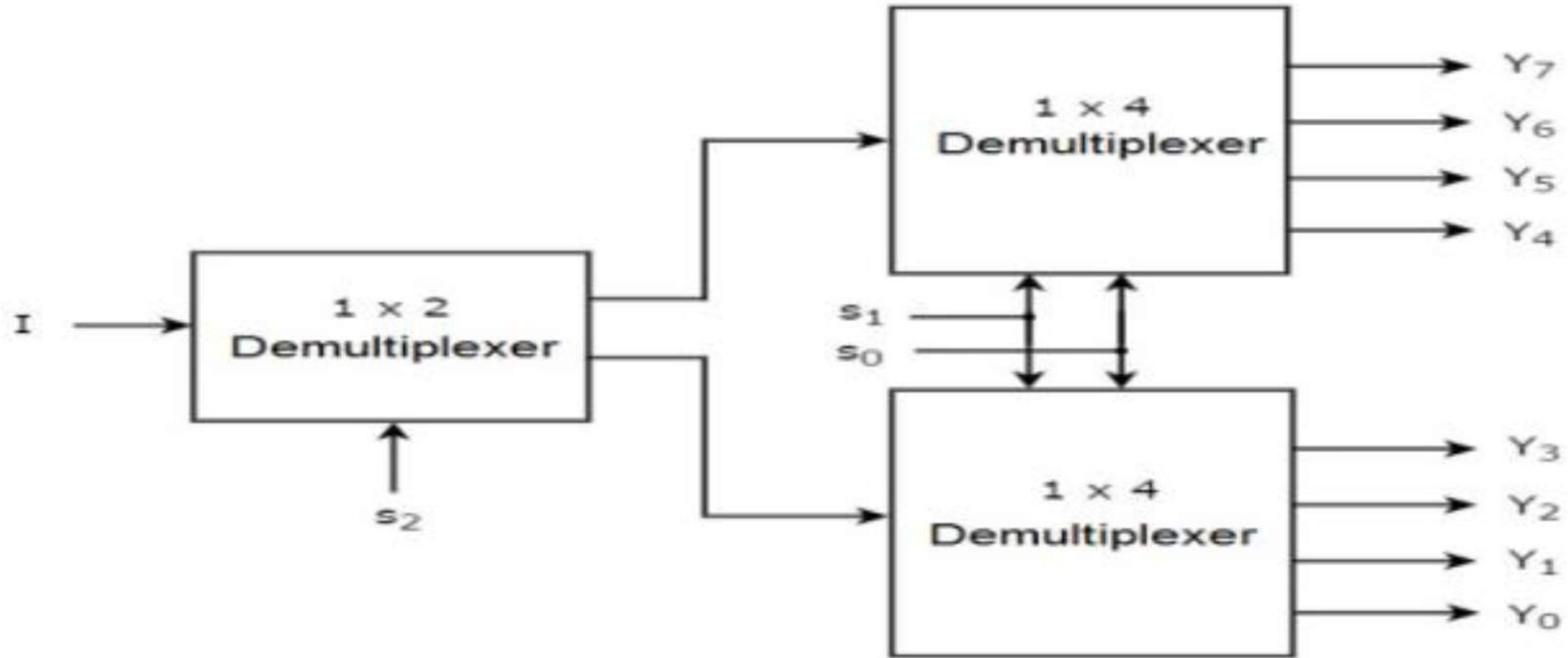




Let the 1×8 Demultiplexer has one input I , three selection lines s_2 , s_1 & s_0 and outputs Y_7 to Y_0 . The **Truth table** of 1×8 Demultiplexer is shown below.

Selection Inputs			Outputs							
s_2	s_1	s_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0





$8/4=2$, $2/4= 0.5$ That is why 1x8 does not create by only 1x4 and it will take help by 1x2 demux as well.



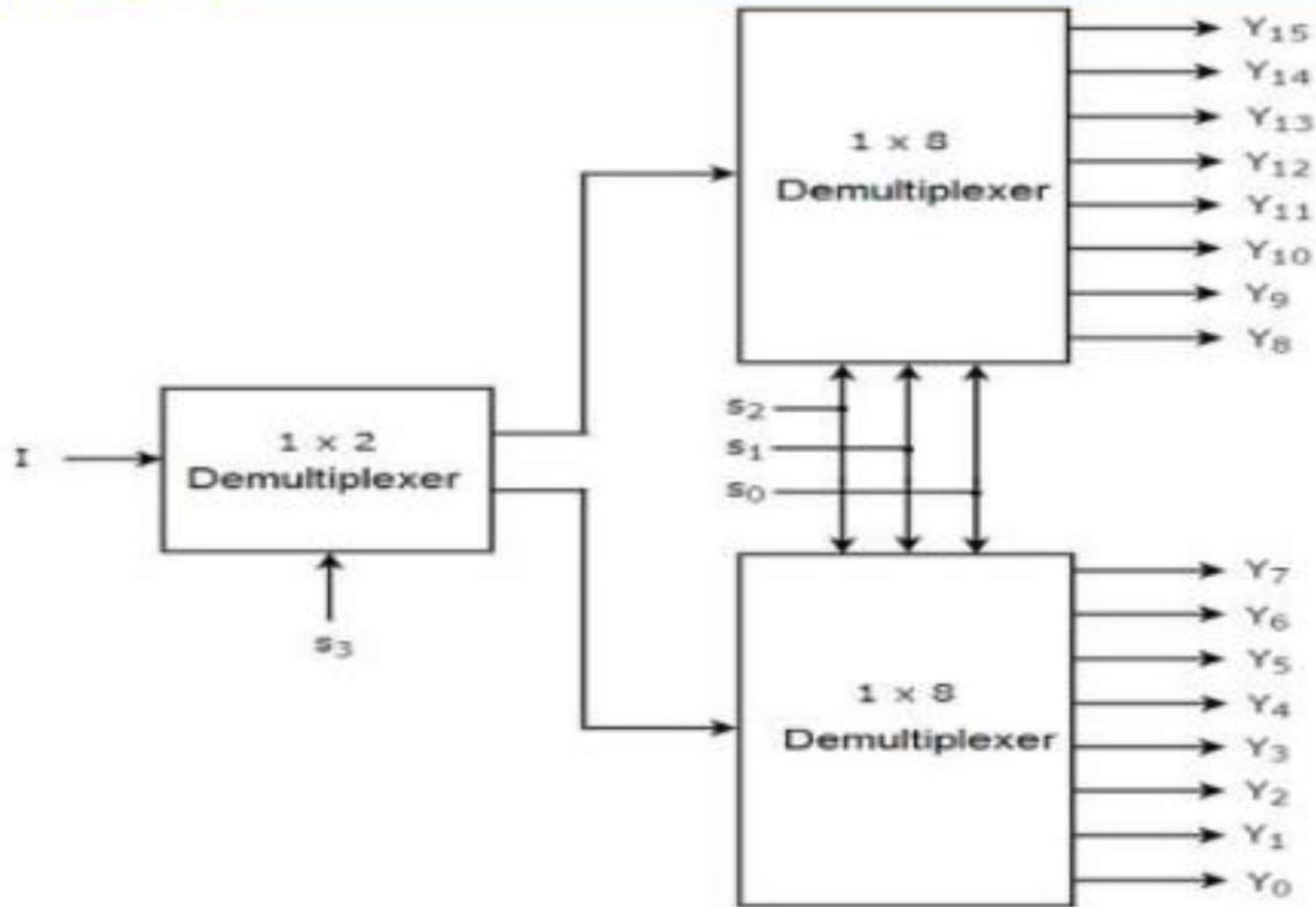
Create it own by you, It is Home-Work



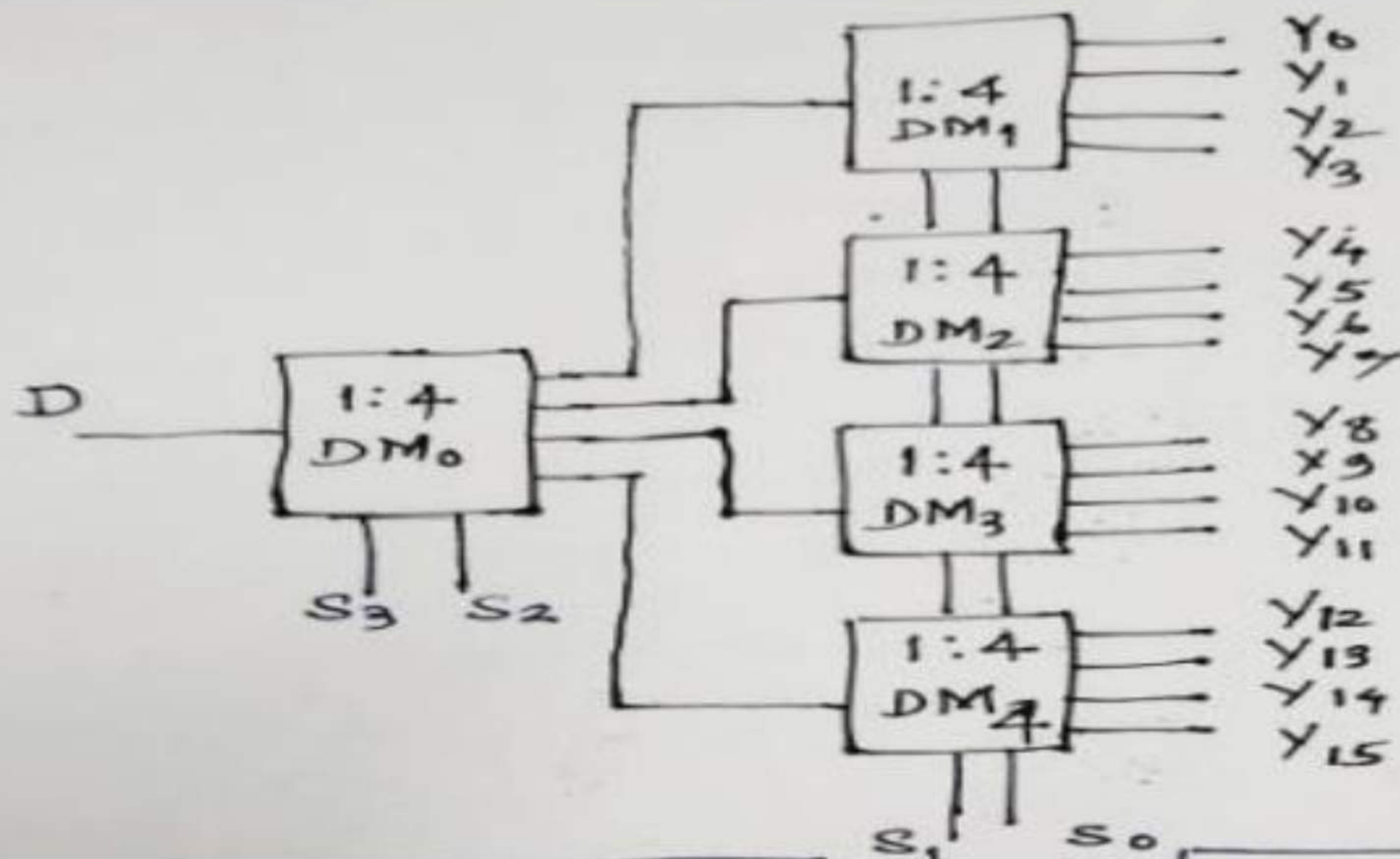
Topic : 1x16 Dmux using 1x8 Demux and 1x2 Demux- Special Case



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$16/8=2$, $2/16= 0.125$ That is why 1x16 does not create by only 1x8 and it will take help by 1x2 demux as well.





Advantages of the DEMUX

- The DEMUX increases the efficiency of the particular communication system as it takes data from a specific input source and distributes it to different sources.
- The DEMUX helps to separate the different signals from the mixed data sources. Then it distributes these data to different sources.
- DEMUX can decode the signal outputs of the [multiplexer](#), as the system works in a reverse way of the MUX.

Disadvantages of the DEMUX

- The DEMUX may cause a wastage of bandwidth as it distributes the refined data in different channels. These channels can overlap with each other which leads to the loss of signal.
- The DEMUX may cause problems in the synchronization of signals. The data channels can overlap with each other which leads to the delay in the whole process.



Applications of DEMUX

- DEMUXs are majorly used to design the ALU circuits and parallel data segments.
- The DEMUXs convert the output of the MUX into the actual input. On the receiver end, the DEMUX can be used to verify the original form of the data to carry out the entire communication process.
- DEMUX helps to save the output. The output generally saved to the ALU. The output will be saved in the registers and the various storage units of the system.
- Each DEMUX has a connection with multiple registers which helps to store the processed data into it.
- The counterpart of the DEMUX regulates the data signal at the output stage of the DEMUX operation. These data can be retrieved later to read out parallelly.
- In the audio or video system, the DEMUX distributes them in different channels. In broadcasting, the DEMUX separates the composite signal into an individual one.

THANK YOU